What if central bank and government fight?

The story of fiscal-monetary conflict and its consequences

based on Bianchi & Melosi (2019)

Warsaw Reading Group

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Growing debt

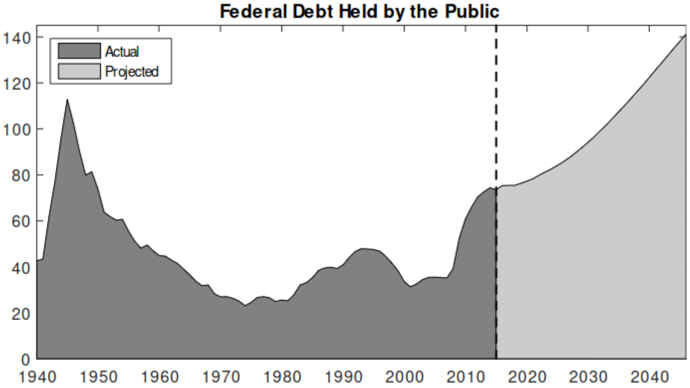


Figure: The planned US debt from 2017 onward

How will that affect the monetary policy?

Endowment economy with government

Let us begin with a simple deterministic model: production $Y_t = Y$ every period, households derive utility from consumption $U(C_t)$ and buy government bonds. The government issues new debt and raises taxes. The budget constraint of the government is:

$$\tau_t P_t + Q_t B_t = B_{t-1},$$

while the Fisher equation (from $C_t = Y$):

$$Q_t^{-1} = \beta^{-1} \frac{P_{t+1}}{P_t}$$

Endowment economy with government

Result:

$$P_t = \frac{\beta P_{t-1} B_{t-1}}{B_{t-2} - P_{t-1} \tau_{t-1}}$$

For determinacy, we just need to find condition for P_1. We can get it from transversality condition!

$$\frac{B_0}{P_1} = \sum_{s=0}^{\infty} \beta^s \tau_s$$

For any given path of (B_t, τ_t) the price level is determinate.

Stochastic environment

Nowe we create more elaborate model, with household problem:

$$\max_{C_t, B_t} \sum_{t=0}^{\infty} \beta^t \exp(\epsilon_t^d) U(C_t)$$

under budget constraint:

$$P_t C_t + \tau_t P_t + Q_t B_t = P_t Y_t + B_{t-1}$$

which gives rise to the Fisher equation:

$$Q_t^{-1} = E_t(\beta^{-1} \frac{\exp(\epsilon_t^d)}{\exp(\epsilon_{t+1}^d)} \frac{P_{t+1}}{P_t})$$

Stochastic environment

With monetary rule:

$$\frac{R_t}{R^*} = (\frac{\Pi_t}{\Pi^*})^{\psi}$$

and government rule:

$$\tau_t - \tau^* = \delta(b_{t-1} - b^*) + \epsilon_t^{\tau}$$

Fiscal and monetary dominance

After linearizing and solving we get:

$$\begin{bmatrix} E_t \tilde{\pi}_{t+1} \\ \hat{b}_t \end{bmatrix} = \begin{bmatrix} \psi & 0 \\ b_* \left(\psi - \beta^{-1}\right) & \beta^{-1} - \delta \end{bmatrix} \begin{bmatrix} \tilde{\pi}_t \\ \hat{b}_{t-1} \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \varepsilon_t^d \\ \varepsilon_t^\tau \end{bmatrix}$$

This has 2 solutions, which are determinate and bounded: (1) $\psi > 1$, $\beta^{-1} - \delta < 1$ and (2) $\psi < 1$, $\beta^{-1} - \delta > 1$.
We call (1) monetary led, and in it inflation is insulated from fiscal shocks. However, in (2), which we call fiscally led, there is no such distinction.

Possible conflict?

The conflict is normally not possible - because variables are either indeterminate or unbounded. However, if conflicts are to be resolved, there exist a way to make them work! This is a central part of this paper.

Introduce 2 other possibilities:

(1) Conflict with a monetary-led resolution - this is easy, the future inflation is only determined by the central bank, not by debt. The stronger central bank response, the higher the debt, the greater future fiscal consolidation and the smaller the inflation.

(2) Conflict with a fiscally-led resolution

Fiscally-led resolution

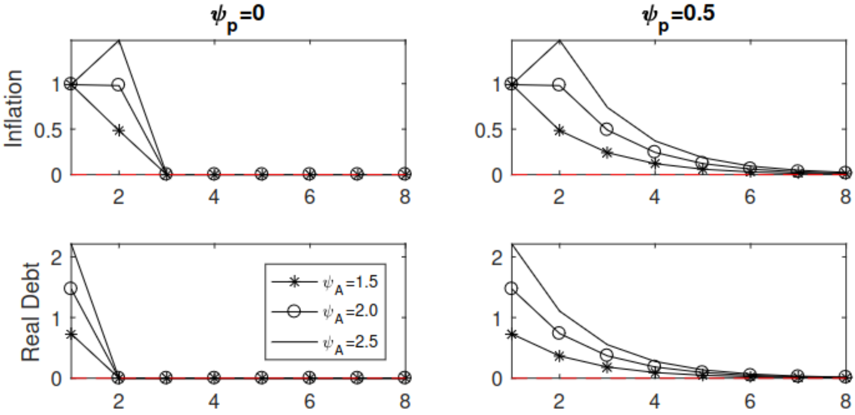


Figure: The effects of different central bank policies

New Keynesian model

Then, a New Keynesian model is created, with Markov process ruling preferences:

$$H^d = \begin{bmatrix} p_{hh} & 1 - p_{ll} \\ 1 - p_{hh} & p_{ll} \end{bmatrix},$$

When demand is high, 3 possibilities arise: monetary dominance, fiscal dominance and conflict. However, when demand is low, we assume that government disregards the level of debt, while central bank deemphasizes inflation stabilization. This assumption is not central to the paper.

All in all, regimes are chosen according to the matrix

$$Q = \begin{bmatrix} p_{hh} Q^H & (1 - p_{ll}) Q^O \\ (1 - p_{hh}) Q^I & p_{ll} Q^L \end{bmatrix},$$

Resolutions

However, if the conflict (active monetary policy/active fiscal policy) regime is expected to eventually end, the model can still admit a stable and unique rational expectations equilibrium. The model could present temporary explosive dynamics, but as long as these are not expected to last for too long, a stationary solution would still exist.

Conflict and fiscally-led resolution

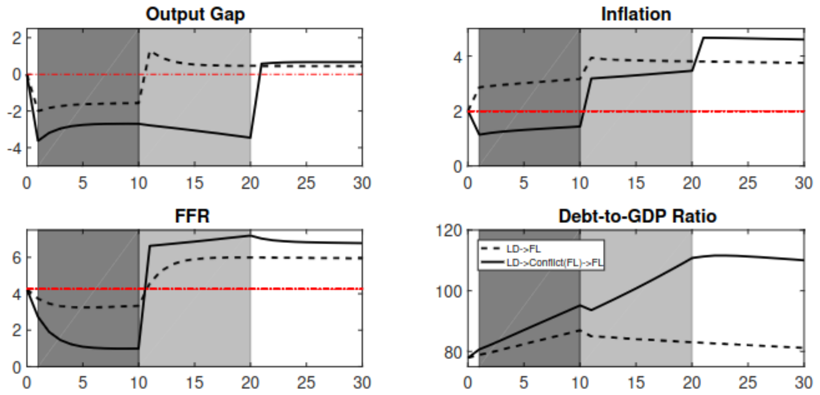


Figure: Comparison of conflict and no-conflict in fiscally-led case

Conflict and monetary-led resolution

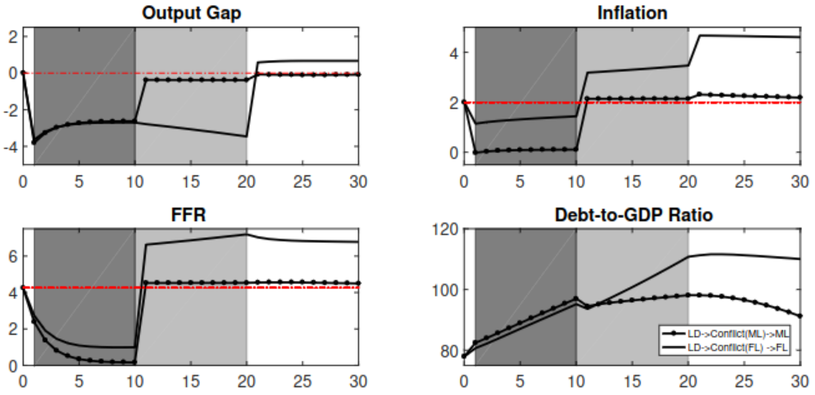
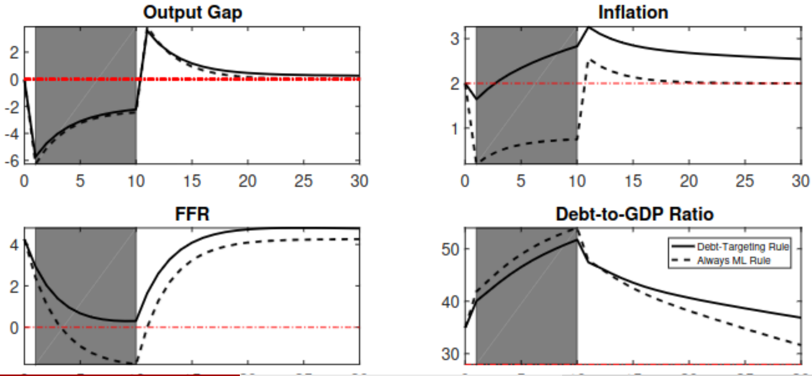


Figure: Comparison of conflict and no-conflict in monetary-led case

Coordination mechanic

We may implement a "shadow" economy, in which no discrete shock happened, and compare it to our model. It is an economy, in which all variables are the same as before the shock, and then we assume the shock (and associated regime change) never happened.



Conclusions

- Maybe monetary policy shocks by 200bp are not rational?
- Central Bank and General Government cannot be fully independent - their actions have influences for each other!
- The conflict is bad for economy - it leads to worse results
- Following the crisis, the agreement may be a better option